

```

In [2]: import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint

def lorenz(state, t, sigma, rho, beta):
    x, y, z = state
    dx_dt = sigma * (y - x)
    dy_dt = x * (rho - z) - y
    dz_dt = x * y - beta * z
    return [dx_dt, dy_dt, dz_dt]

t = np.linspace(0, 50, 10000)

params = [
    (14, 10, 8/3),
    (15, 10, 8/3),
    (28, 10, 8/3)
]

# Initial conditions
initial_conditions = [
    [1.0, 1.0, 1.0],
    [1.01, 1.0, 1.0]
]

for rho, sigma, beta in params:
    for ic in initial_conditions:

        solution = odeint(lorenz, ic, t, args=(sigma, rho, beta))
        x, y, z = solution.T

        plt.figure(figsize=(15, 5))

        plt.subplot(131)
        plt.plot(x, y)
        plt.title(f'x vs y (ρ={rho}, σ={sigma}, β={beta})')
        plt.xlabel('x')
        plt.ylabel('y')

        plt.subplot(132)
        plt.plot(y, z)
        plt.title(f'y vs z (ρ={rho}, σ={sigma}, β={beta})')
        plt.xlabel('y')
        plt.ylabel('z')

        plt.subplot(133)
        plt.plot(x, z)
        plt.title(f'x vs z (ρ={rho}, σ={sigma}, β={beta})')
        plt.xlabel('x')
        plt.ylabel('z')
        plt.tight_layout()
        plt.show()

    fig = plt.figure(figsize=(10, 8))

```

```

ax = fig.add_subplot(111, projection='3d')
ax.plot(x, y, z, lw=0.5)
ax.set_title(f'3D Phase Space ( $\rho$ = $\rho$ ,  $\sigma$ = $\sigma$ ,  $\beta$ = $\beta$ )')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('z')
plt.show()

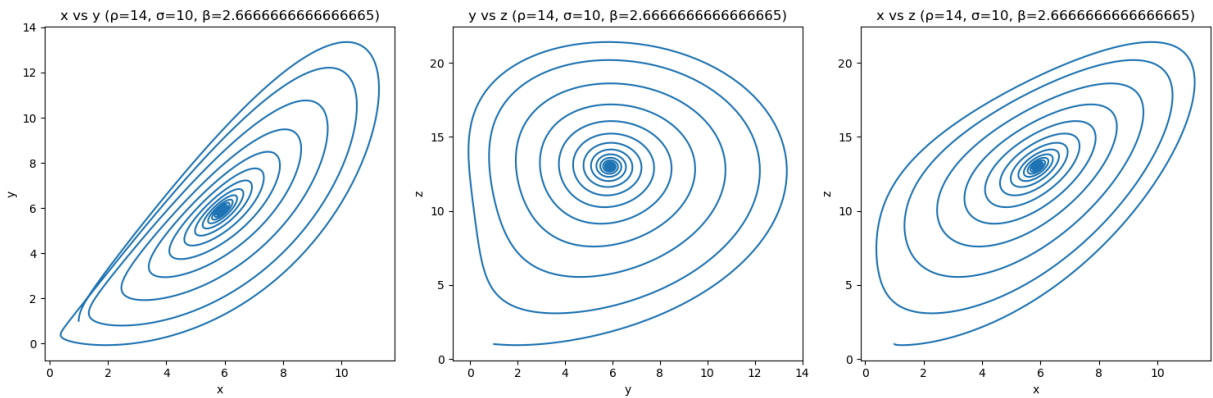
plt.figure(figsize=(15, 5))

plt.subplot(131)
plt.plot(t, x)
plt.title(f'x vs t ( $\rho$ = $\rho$ ,  $\sigma$ = $\sigma$ ,  $\beta$ = $\beta$ )')
plt.xlabel('t')
plt.ylabel('x')

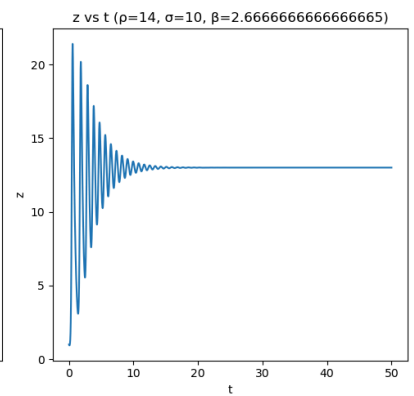
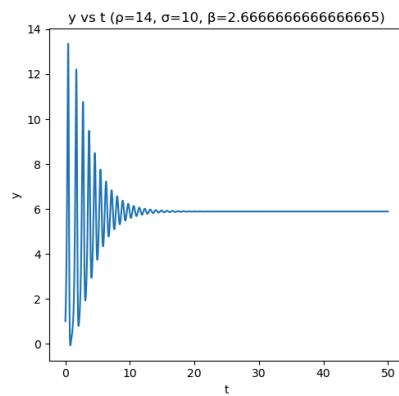
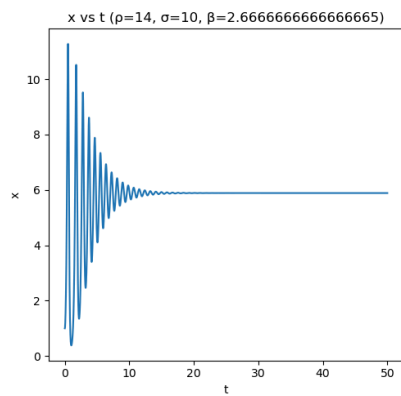
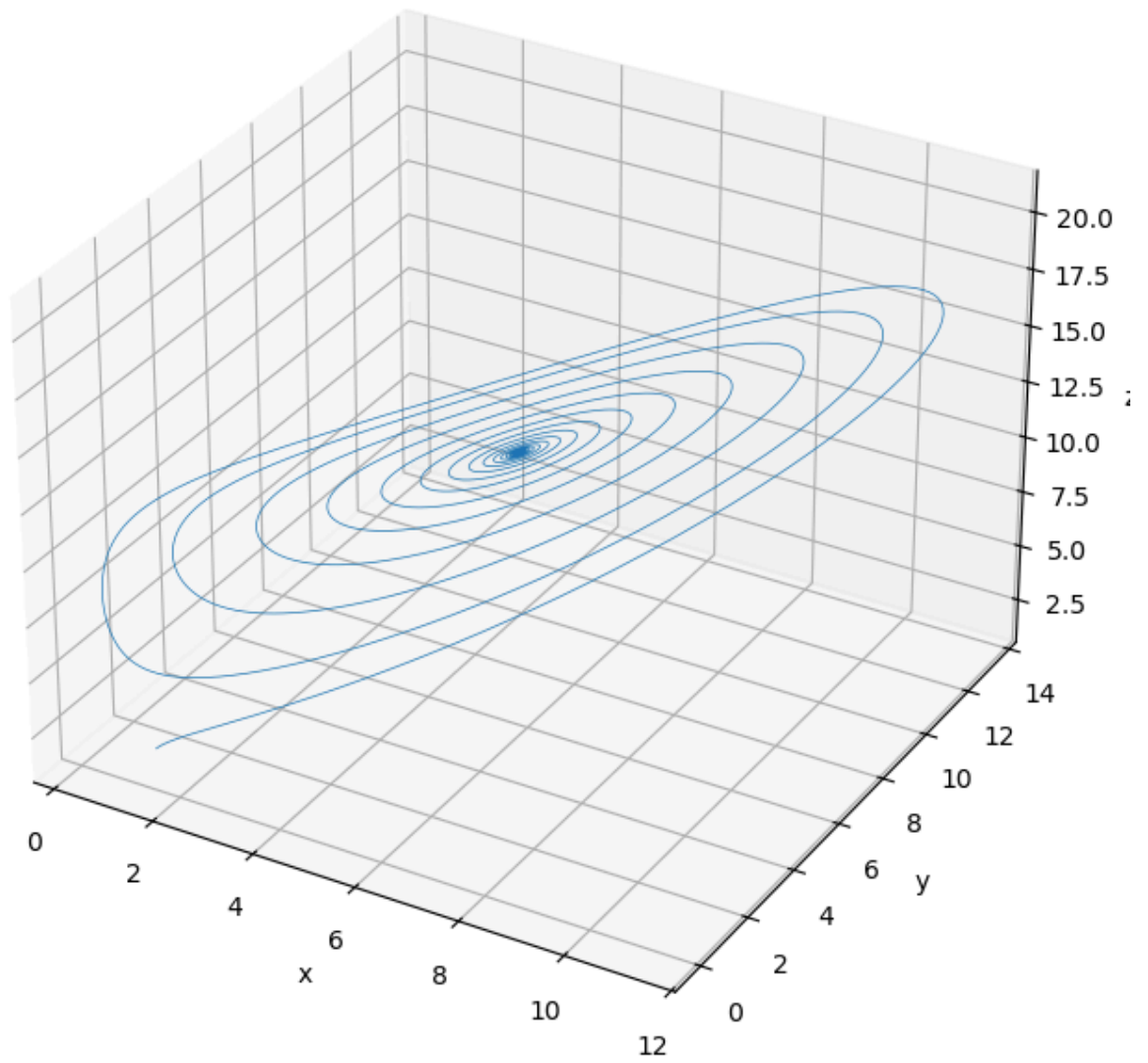
plt.subplot(132)
plt.plot(t, y)
plt.title(f'y vs t ( $\rho$ = $\rho$ ,  $\sigma$ = $\sigma$ ,  $\beta$ = $\beta$ )')
plt.xlabel('t')
plt.ylabel('y')

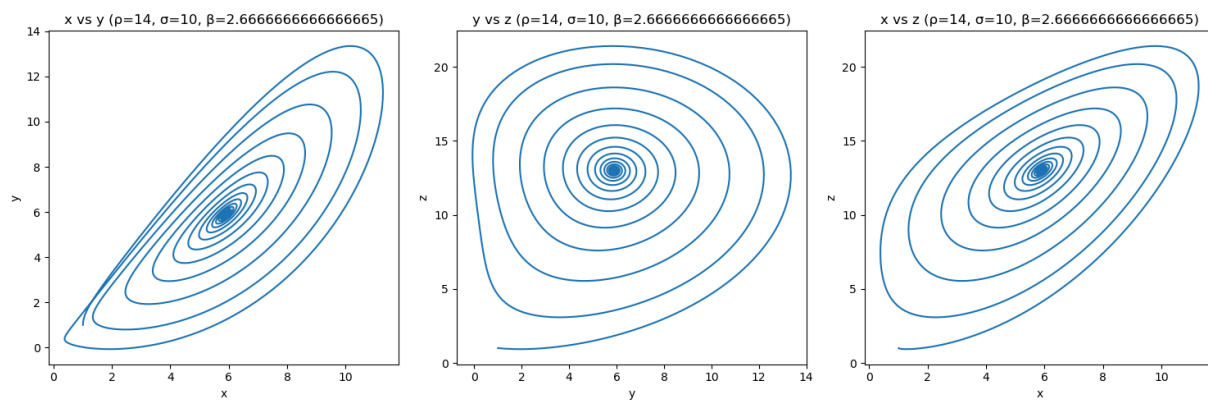
plt.subplot(133)
plt.plot(t, z)
plt.title(f'z vs t ( $\rho$ = $\rho$ ,  $\sigma$ = $\sigma$ ,  $\beta$ = $\beta$ )')
plt.xlabel('t')
plt.ylabel('z')
plt.tight_layout()
plt.show()

```

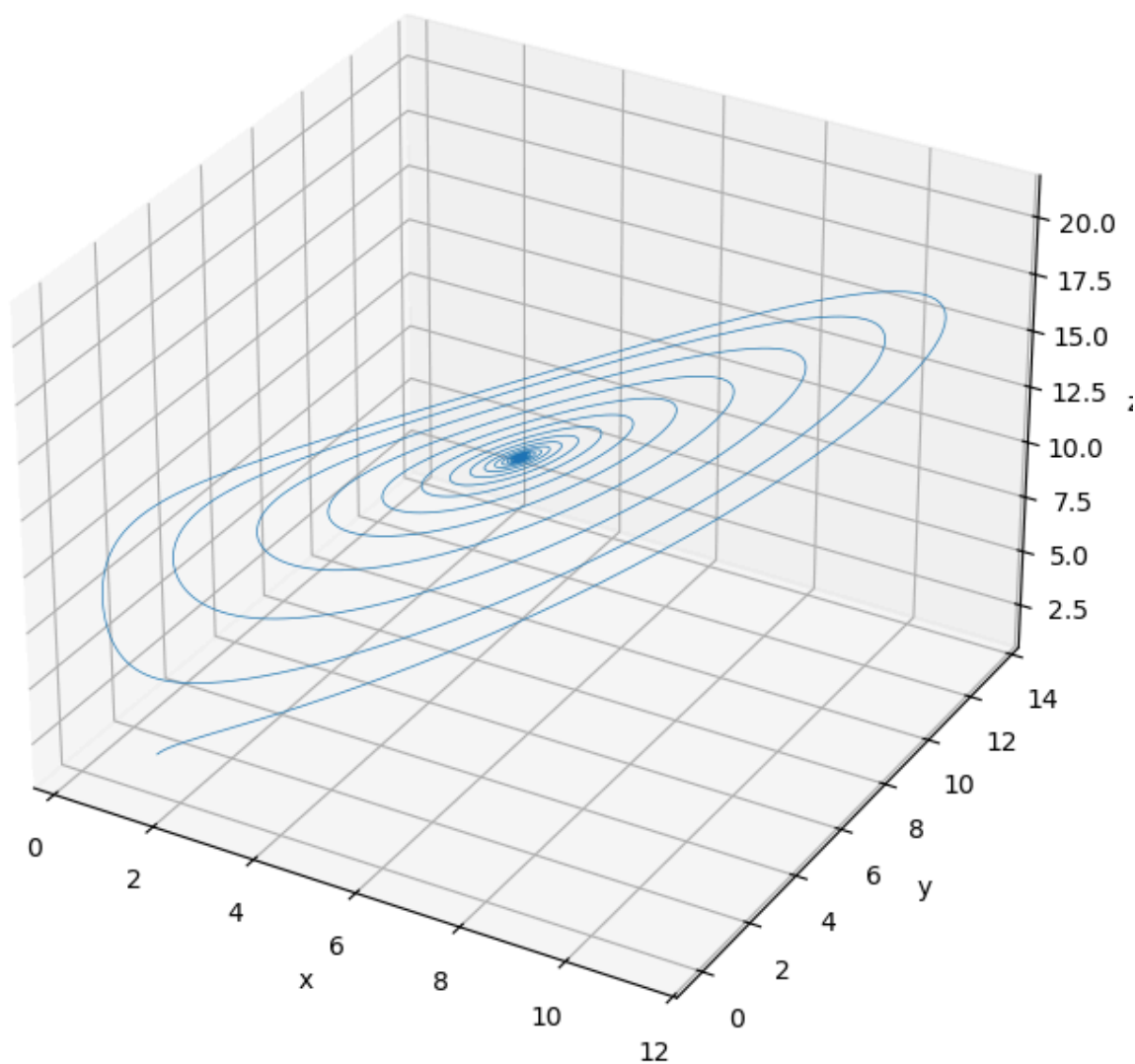


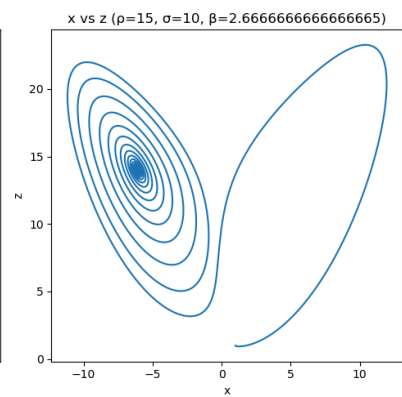
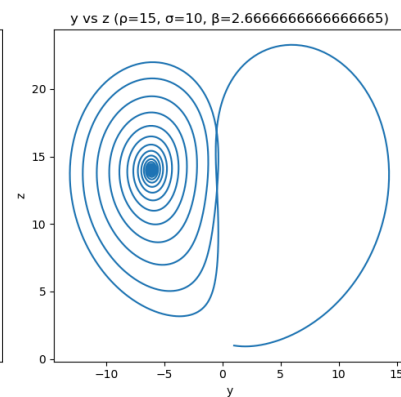
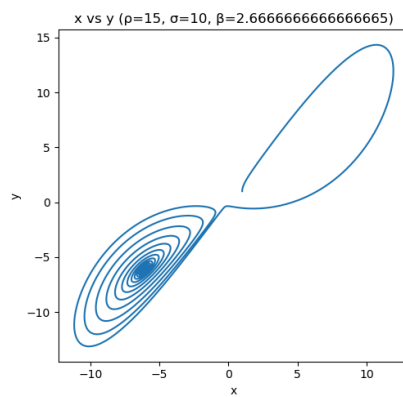
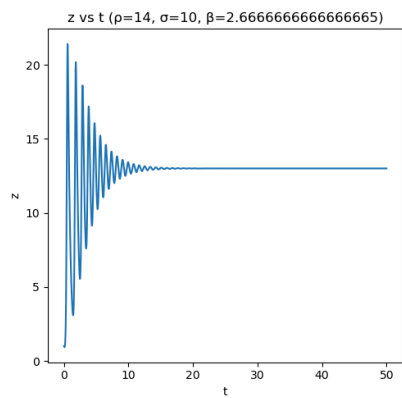
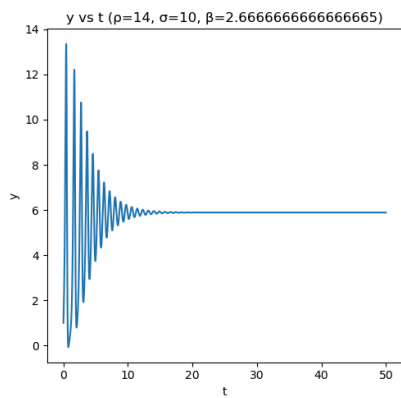
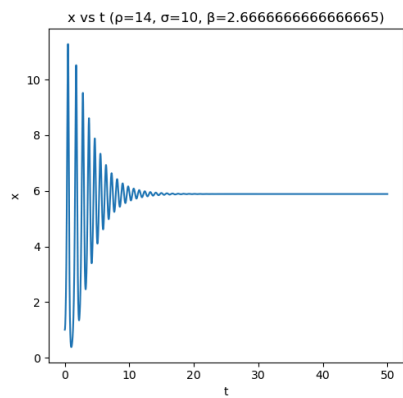
### 3D Phase Space ( $\rho=14$ , $\sigma=10$ , $\beta=2.6666666666666665$ )



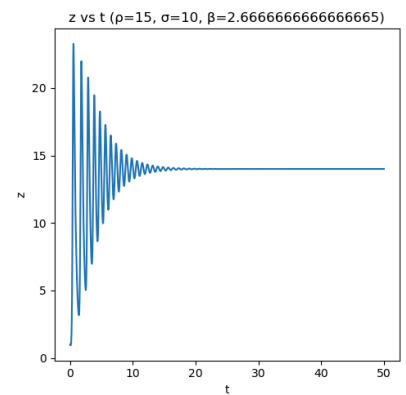
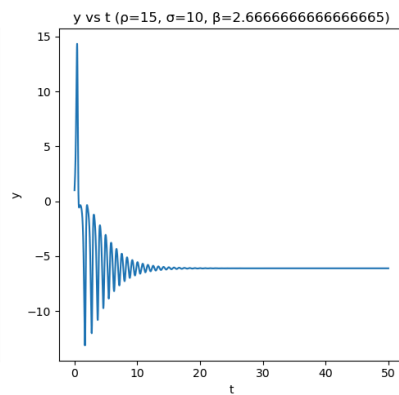
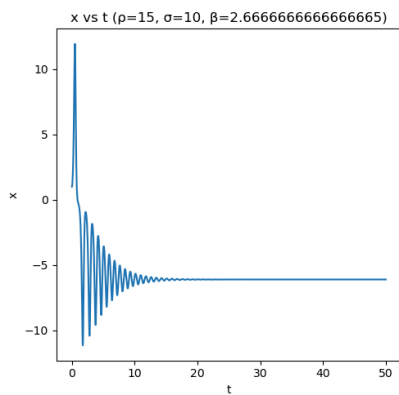
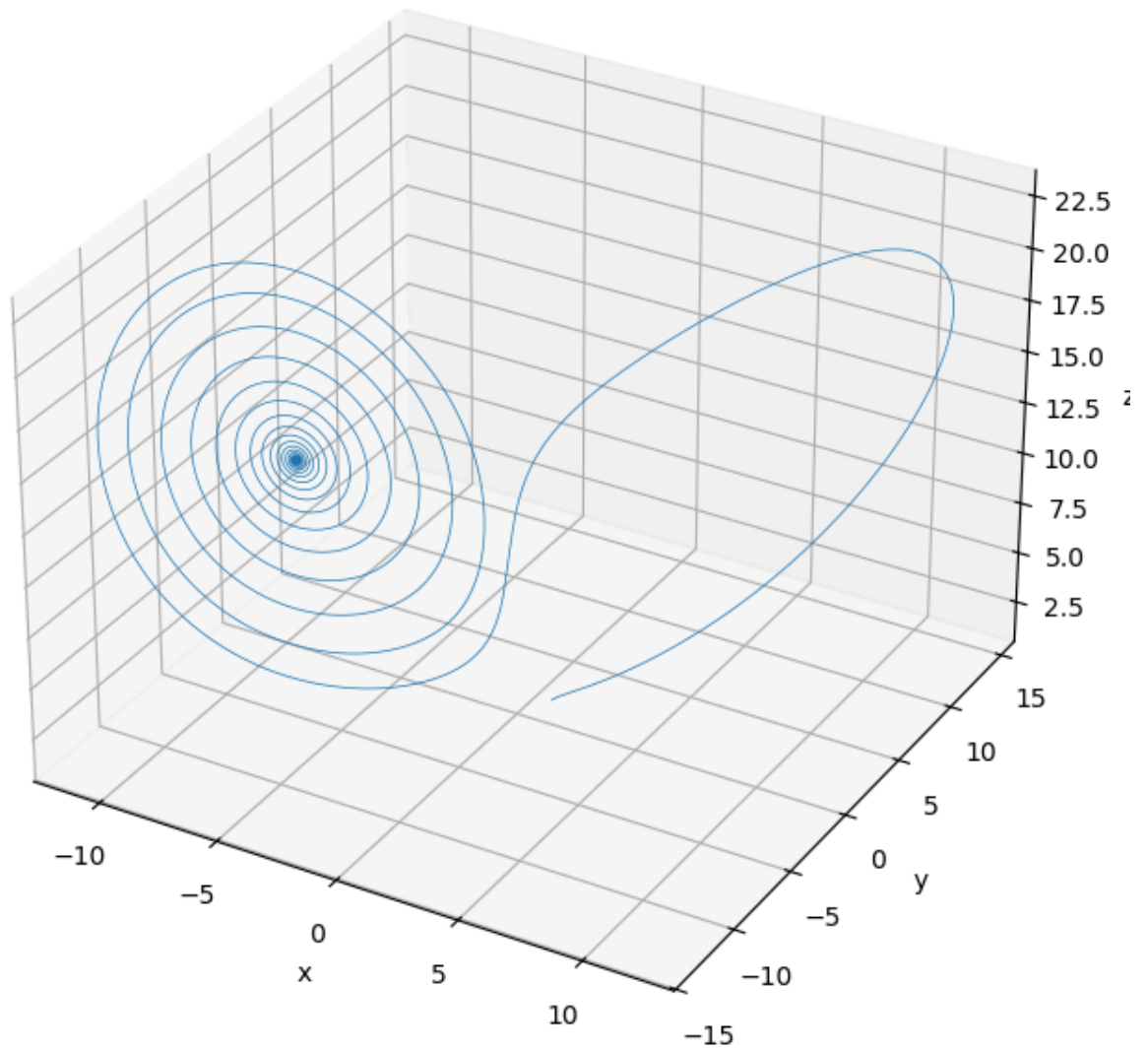


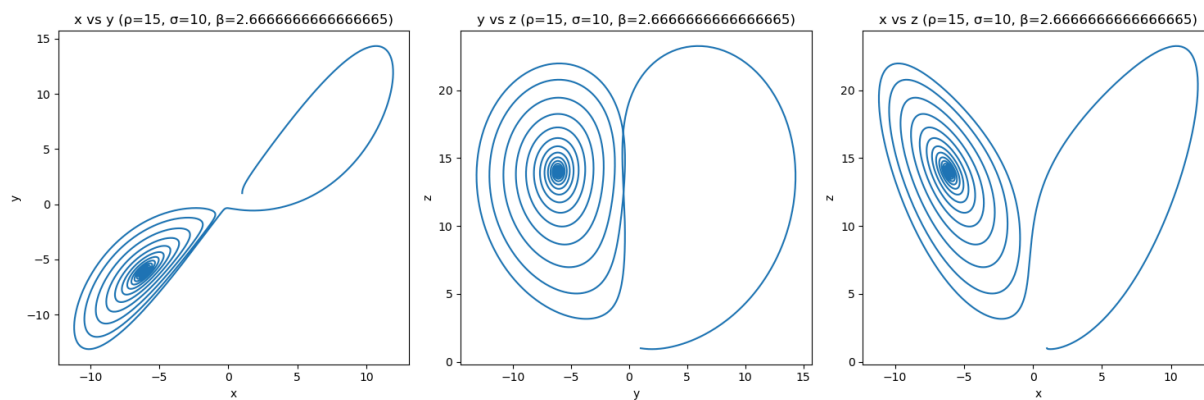
3D Phase Space ( $\rho=14, \sigma=10, \beta=2.666666666666665$ )



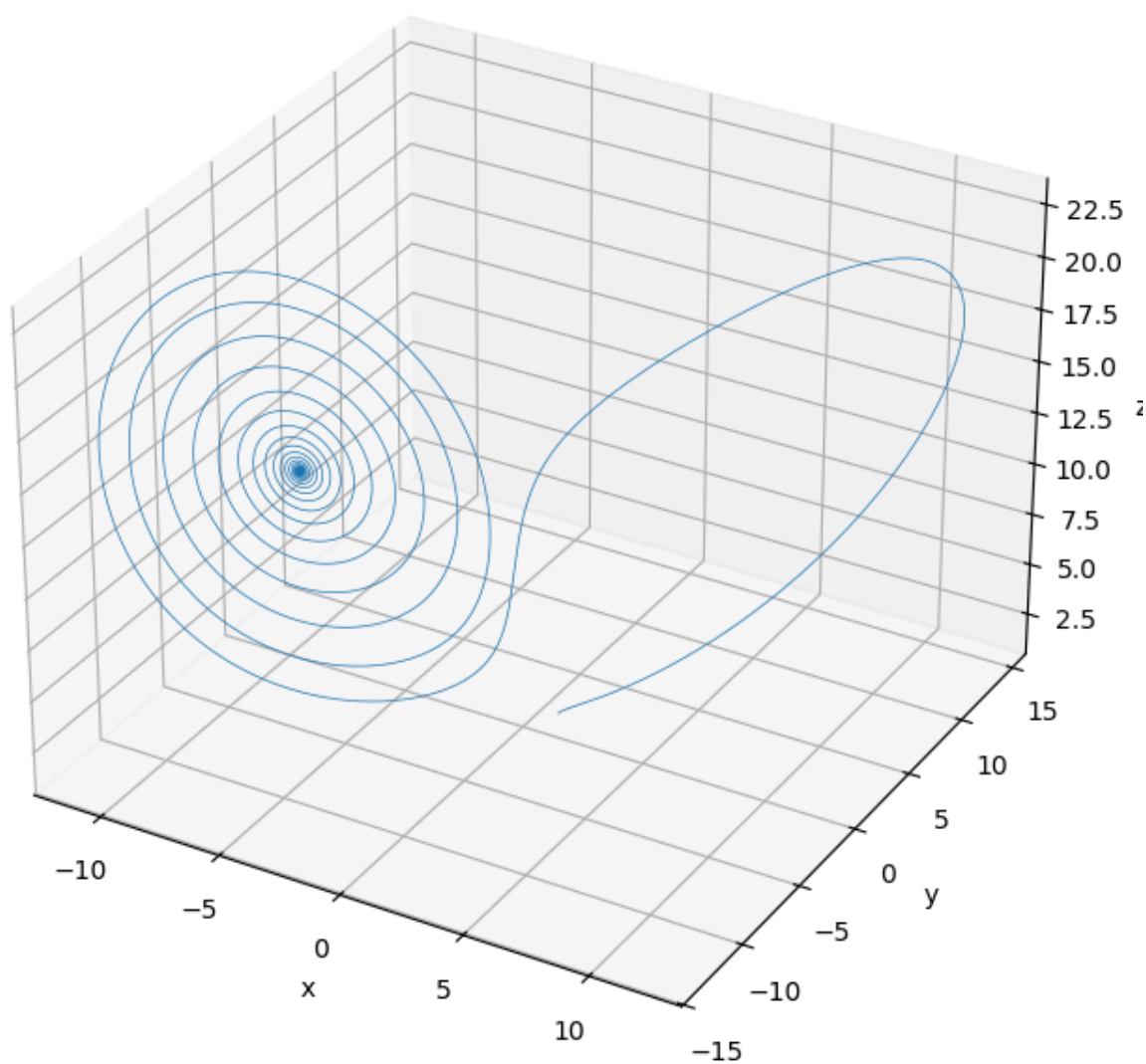


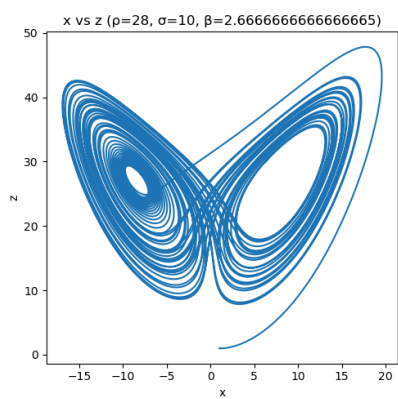
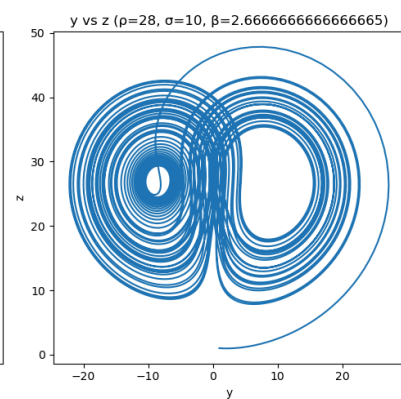
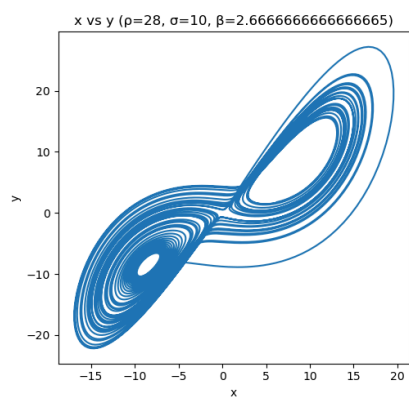
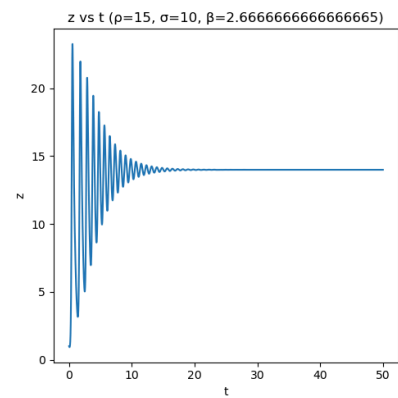
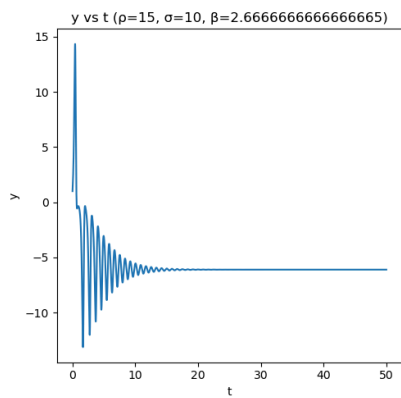
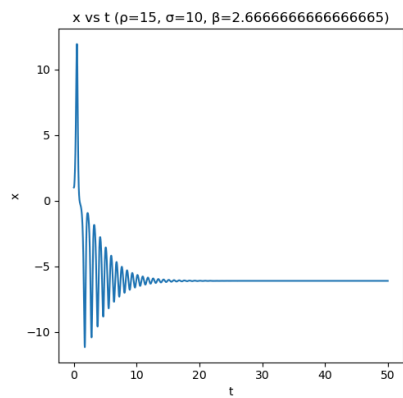
### 3D Phase Space ( $\rho=15$ , $\sigma=10$ , $\beta=2.6666666666666665$ )





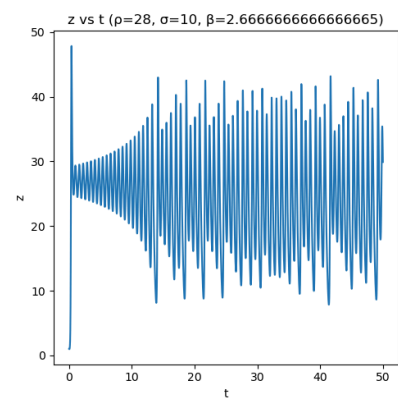
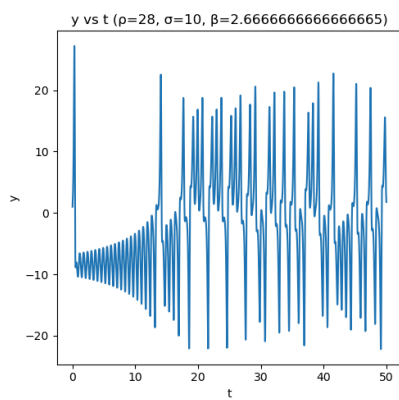
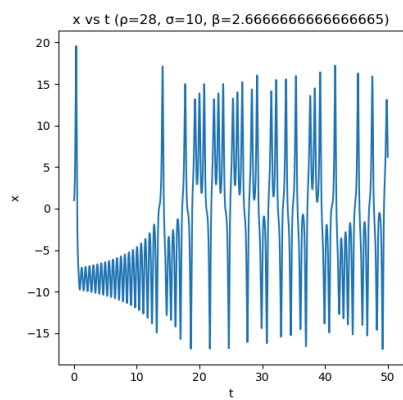
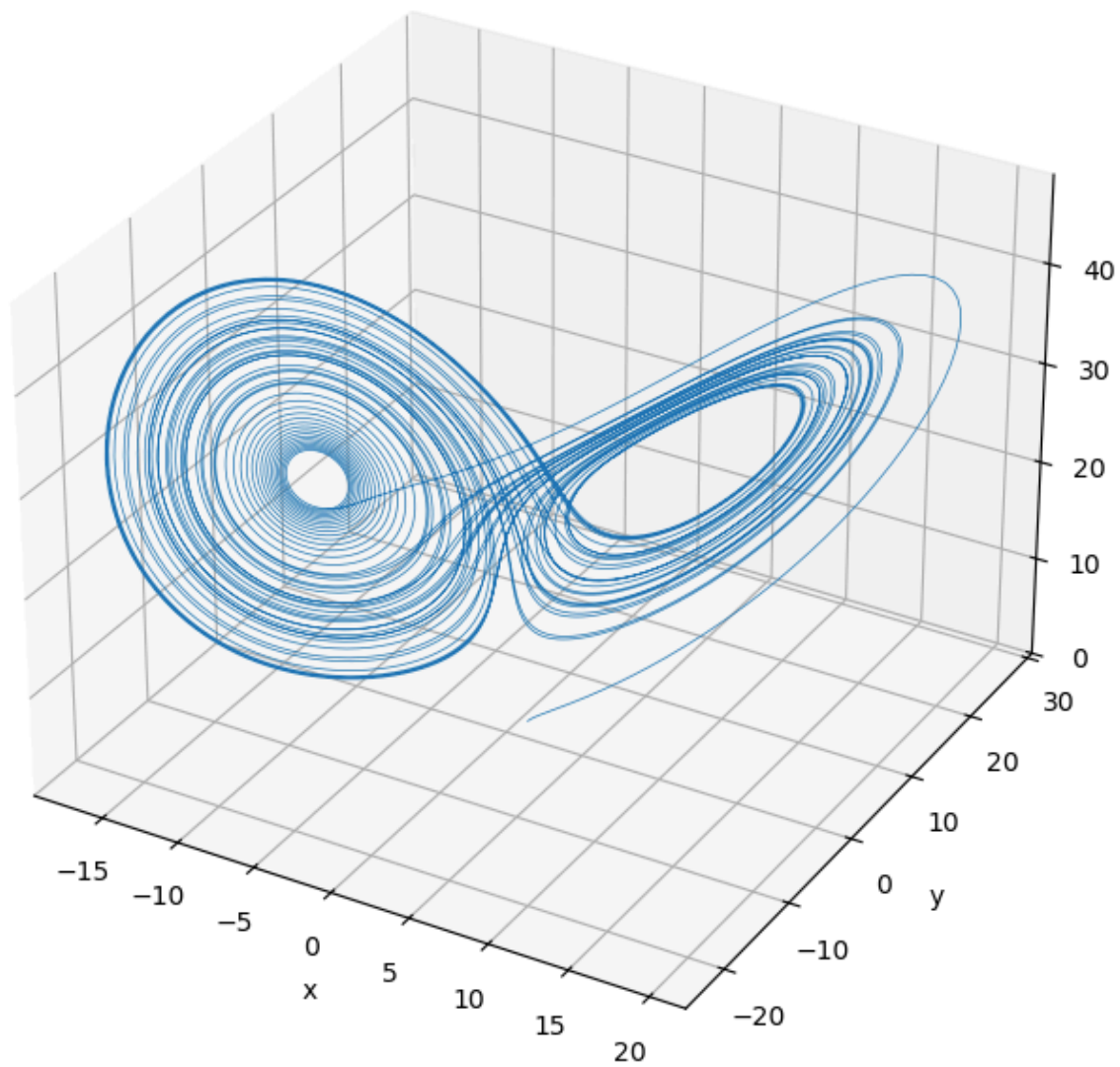
3D Phase Space ( $\rho=15, \sigma=10, \beta=2.6666666666666665$ )

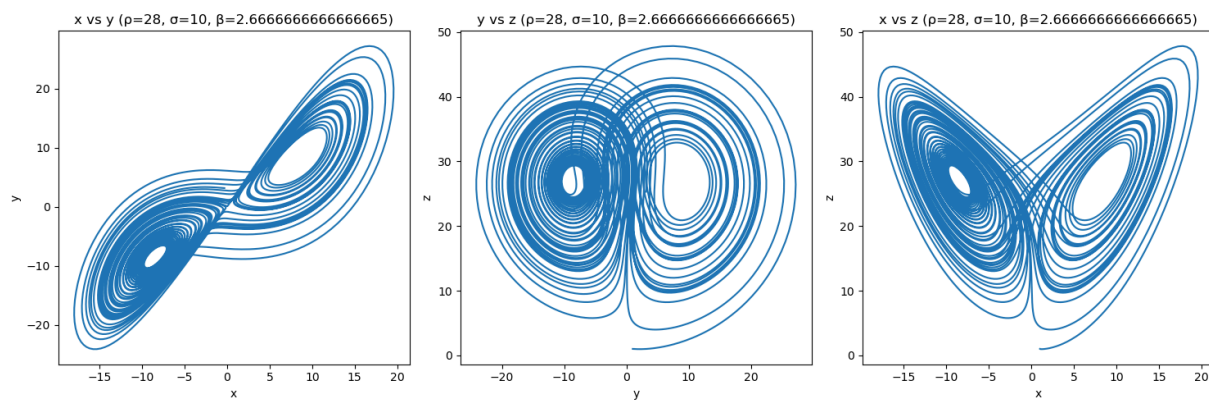




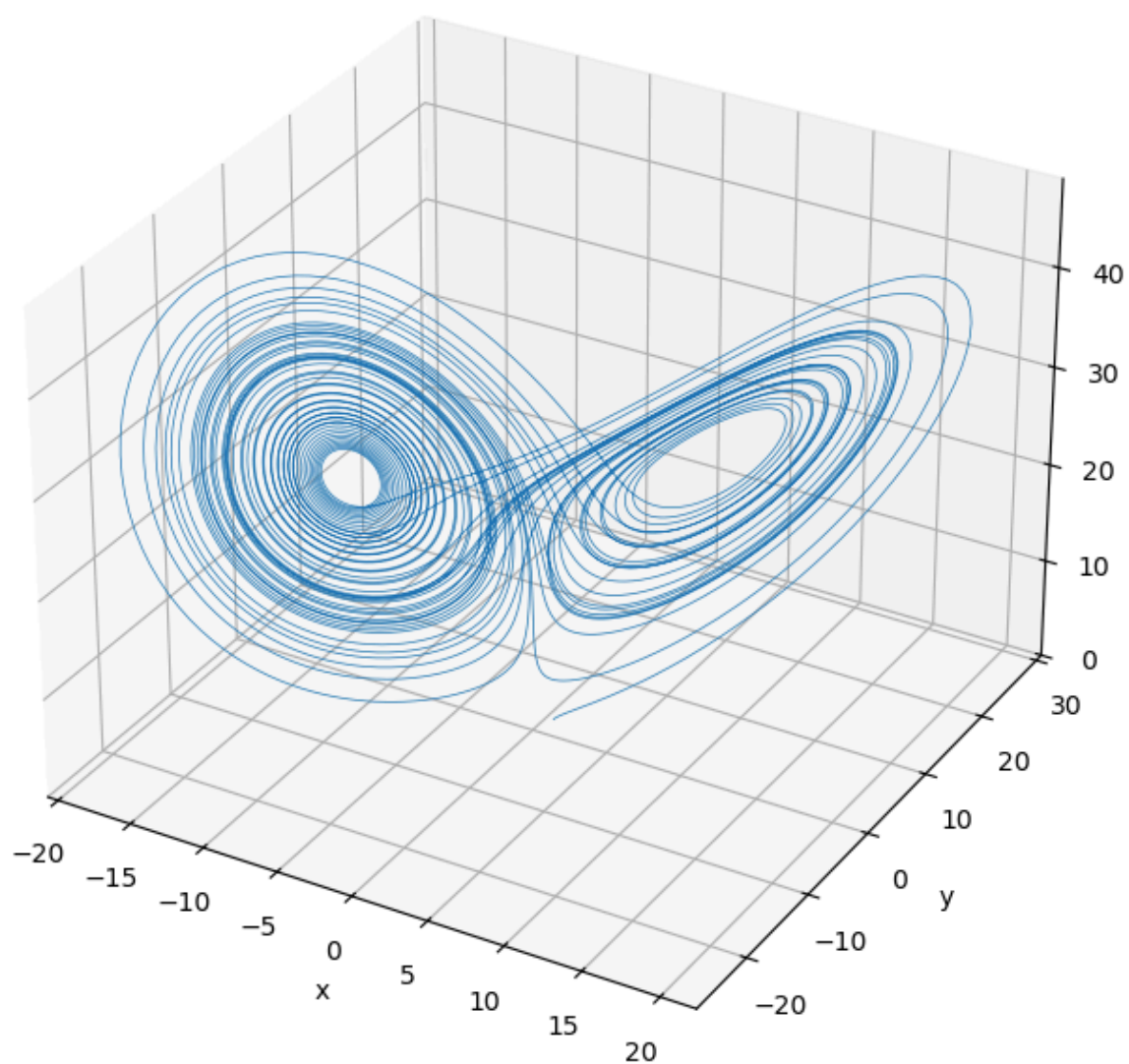


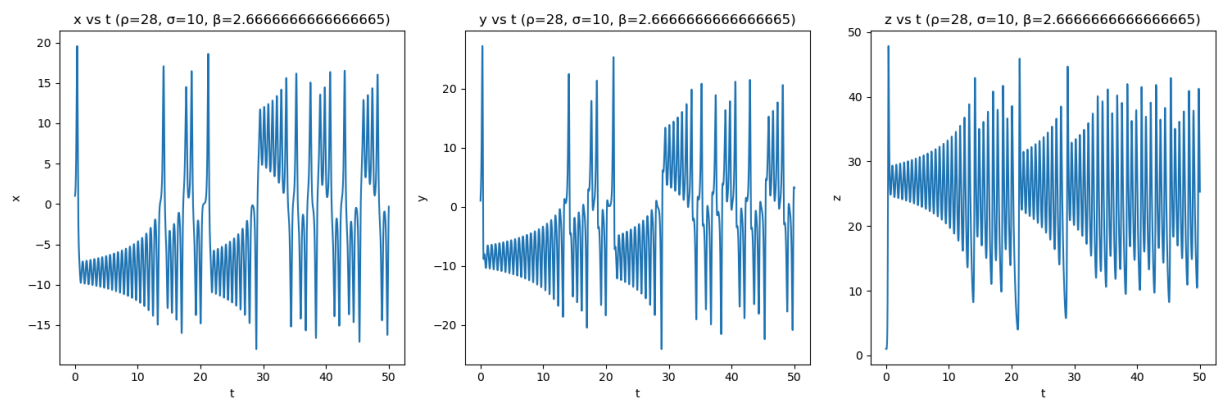
3D Phase Space ( $\rho=28$ ,  $\sigma=10$ ,  $\beta=2.6666666666666665$ )





3D Phase Space ( $\rho=28, \sigma=10, \beta=2.6666666666666665$ )





In [ ]: